

Evolution of Microprocessors

A Journey Through Computing History

| The Dawn of Microprocessing

Foundational Tech

The journey began with the invention of the **Transistor** in 1948, followed by the **Integrated Circuit (IC)** in 1958 by Jack Kilby and Robert Noyce.

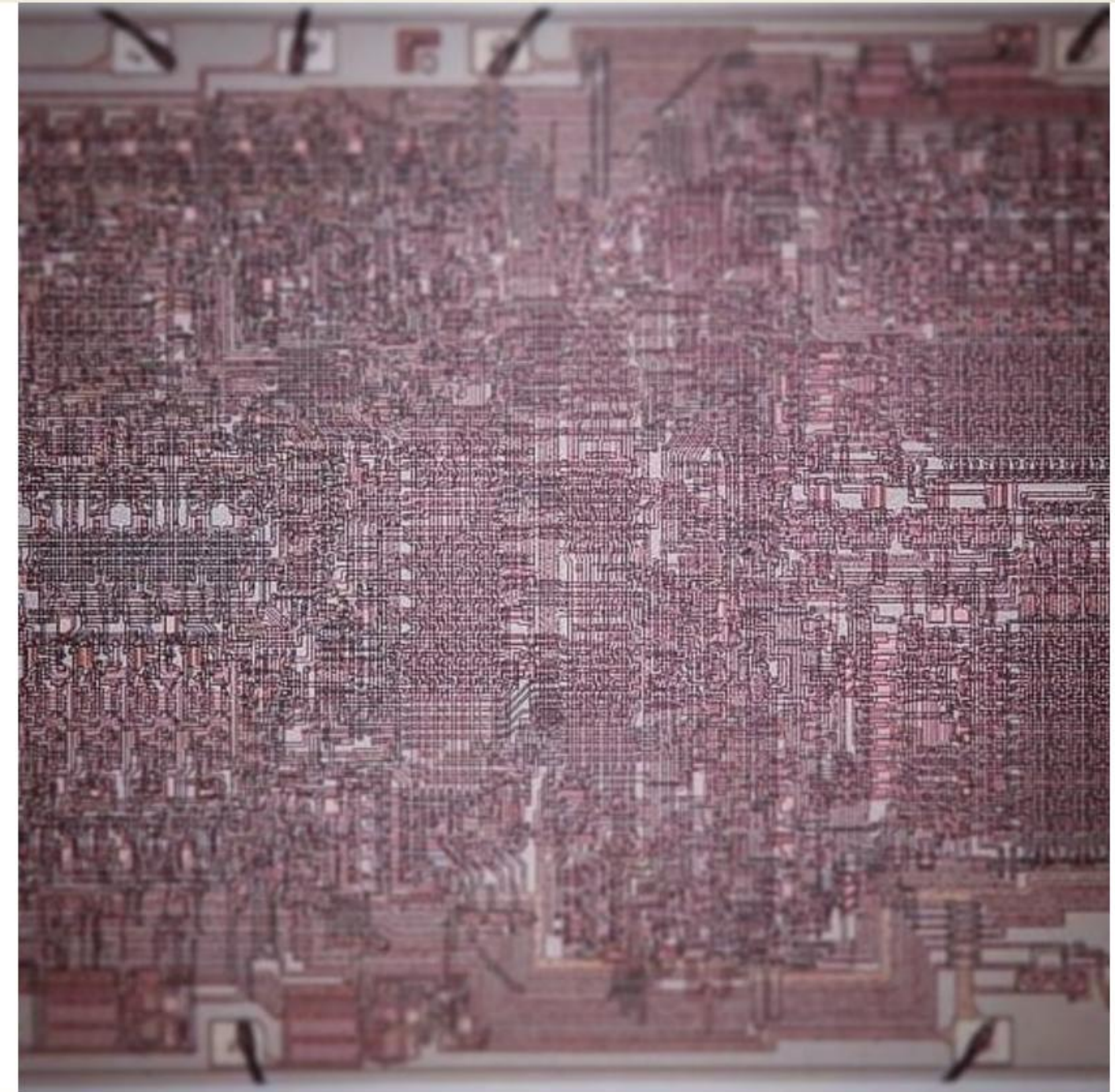
What is a Microprocessor?

A multi-purpose, clock-driven, register-based, digital integrated circuit that accepts binary data, processes it, and provides output.

| 1st Gen: The 4-Bit Era (1971–1973)

Intel introduced the first microprocessor, the **Intel 4004**, in 1971.

- Processing Speed: 740 kHz
- Addressable memory: 640 bytes
- Designated for calculator use primarily.
- Contained 2,300 transistors.

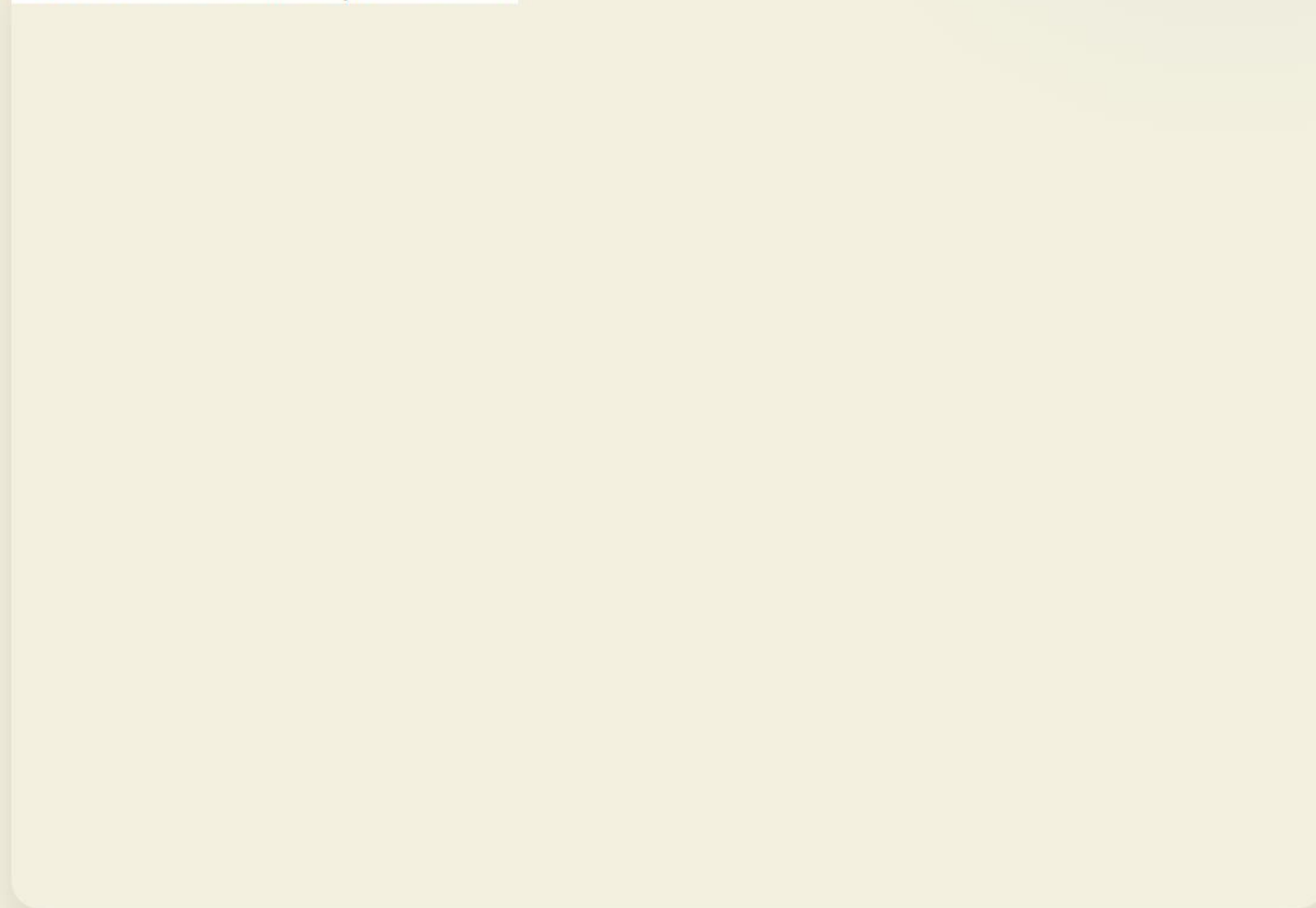


| 2nd Gen: The 8-Bit Era (1974–1978)

This generation saw the rise of legendary chips like the **Intel 8008, 8080, and 8085**.

- 8-bit data bus and 16-bit address bus.
- The 8085 became a staple in academic learning.
- Used in early personal computers and industrial controllers.
- Increased clock speeds up to 3 MHz.

 Intel 8085 Microprocessor



| Expansion: 16-Bit and 32-Bit Eras

3rd Gen (1979–1980)

Introduction of **16-bit** processors like the **Intel 8086** and 8088. This era marked the birth of the IBM PC.

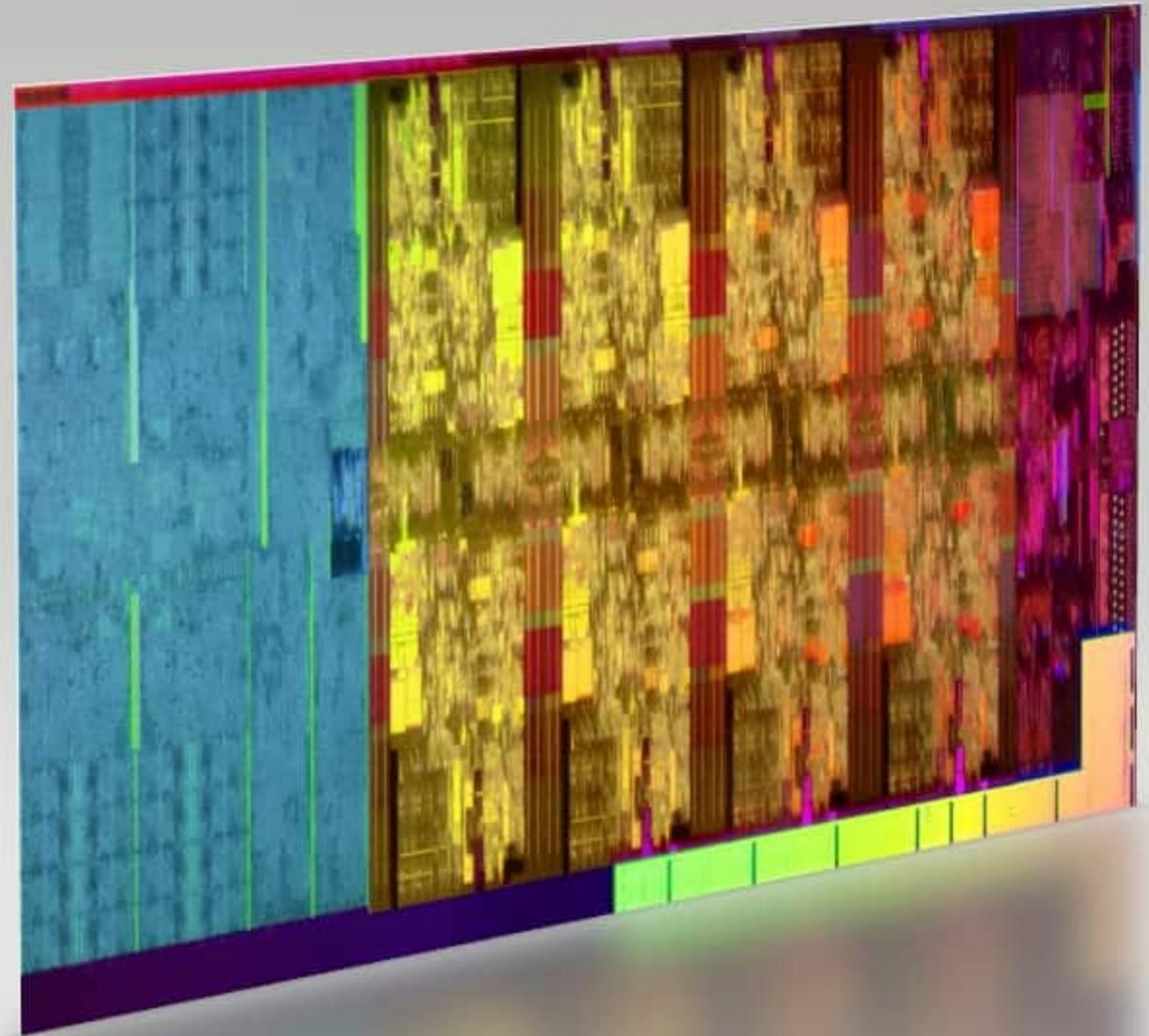
4th Gen (1981–1995)

The leap to **32-bit** architecture with the **Intel 80386, 80486**, and the first **Pentium** chips, reaching speeds of 100MHz+.

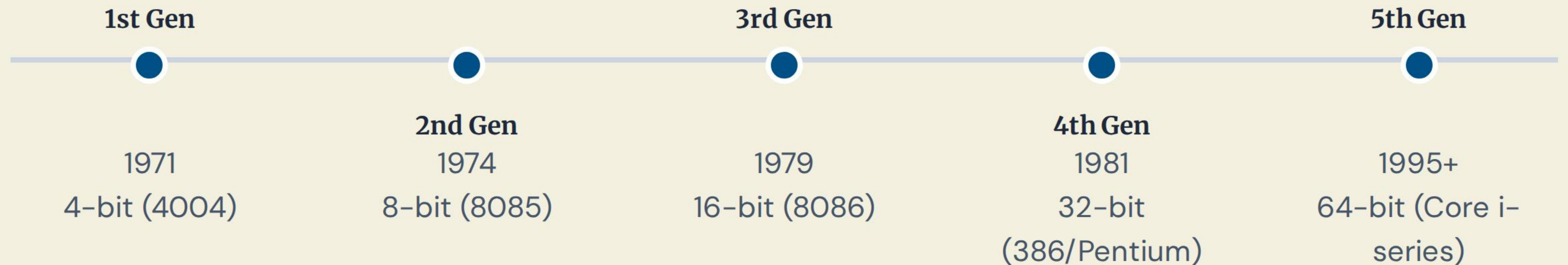
Modern Era: 64-Bit & Multi-Core

Since 1995, we have entered the 64-bit era characterized by parallel processing.

- **Intel Core i3, i5, i7, i9** series.
- **Multi-core architecture:** Dual, Quad, Octa-core.
- Speeds exceeding 5 GHz.
- Billions of transistors on a single die.



Evolutionary Roadmap



| Core Architecture Types



CISC

Complex Instruction Set
Computer. Focus on
hardware-level complexity
(e.g., Intel x86).



RISC

Reduced Instruction Set
Computer. Simple
instructions, faster execution
(e.g., ARM, Apple M1).



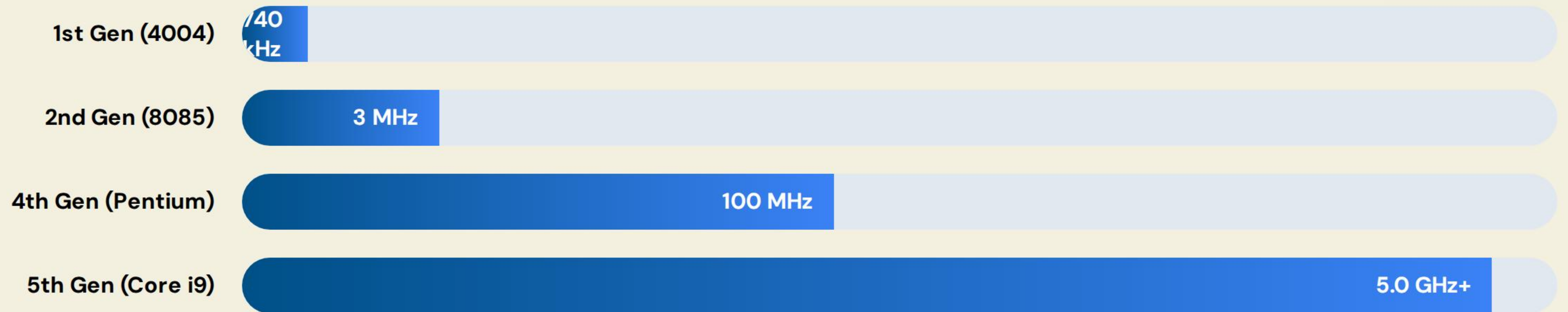
Specialized

ASICs, DSPs, and Superscalar
units designed for niche
tasks like graphics or AI.

| Modern Processor Features

-  **Integrated Cache:** L1, L2, and L3 caches for near-instant data access.
 -  **Built-in Graphics:** Many modern CPUs include an Integrated GPU (iGPU).
 -  **AI Engines:** Specialized neural processing units (NPU) for machine learning tasks.
 -  **Power Management:** Dynamic frequency scaling to save energy and reduce heat.
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| The Exponential Growth of Speed



Note: Scale is non-linear to illustrate the massive jump from kHz to GHz.

Microprocessor Assessment

Factor	Advantages	Limitations
Size & Form	Compact and portable	Board size > Microcontrollers
Performance	Incredible processing speed	Excessive heat from high use
Flexibility	Programmable for any task	Requires external peripherals
Reliability	Extremely low failure rates	Sensitive to voltage fluctuations

Q&A

Thank you for exploring the evolution of microprocessors!

 contact@example.com

 www.geeksforgeeks.org

| Image Sources



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